In [10]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

IMPORT DATA

In [11]:
```
df = pd.read_excel("C:/Users/strip/Desktop/swiggy_data.xlsx")
```

In [4]:
```
df.head()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[4], line 1
----> 1 df.head()

NameError: name 'df' is not defined
```

In [19]:
```
df.tail()
```

Out[19]:

	State	City	Order Date	Restaurant Name	Location	Category	Dish Name	Price (INR)	Rati
197425	Sikkim	Gangtok	2025-01-25	Mama's Kitchen	Gangtok	Momos	Soya cheese chilli momo ...	112.0	4
197426	Sikkim	Gangtok	2025-07-02	Mama's Kitchen	Gangtok	Momos	Kurkure momo fried ...	140.0	4
197427	Sikkim	Gangtok	2025-03-25	Mama's Kitchen	Gangtok	Momos	Chilli cheese momo	126.0	4
197428	Sikkim	Gangtok	2025-03-26	Mama's Kitchen	Gangtok	Momos	Veg Momos (8 Pc)	85.0	4
197429	Sikkim	Gangtok	2025-03-27	Mama's Kitchen	Gangtok	Momos	Soya Momo	100.0	4

MetaData

In [20]:
```
df.shape[0]
```

Out[20]: 197430

In [21]:
```
print("No of Rows:",df.shape[0])
```

```
No of Rows: 197430
```

In [22]:
```
print("No of Fields;",df.shape[1])
```

```
No of Fields; 10
```

In [23]:
```
df.info
```

Out[23]:
```
<bound method DataFrame.info of             State       City Order Date
Restaurant Name  \
0       Karnataka  Bengaluru 2025-06-29  Anand Sweets & Savouries
1       Karnataka  Bengaluru 2025-04-03     Srinidhi Sagar Deluxe
2       Karnataka  Bengaluru 2025-01-15     Srinidhi Sagar Deluxe
3       Karnataka  Bengaluru 2025-04-17     Srinidhi Sagar Deluxe
4       Karnataka  Bengaluru 2025-03-13     Srinidhi Sagar Deluxe
...           ...        ...        ...                       ...
197425     Sikkim    Gangtok 2025-01-25            Mama's Kitchen
197426     Sikkim    Gangtok 2025-07-02            Mama's Kitchen
197427     Sikkim    Gangtok 2025-03-25            Mama's Kitchen
197428     Sikkim    Gangtok 2025-03-26            Mama's Kitchen
197429     Sikkim    Gangtok 2025-03-27            Mama's Kitchen

                    Location     Category  \
0       Rajarajeshwari Nagar        Snack
1                    Kengeri  Recommended
2                    Kengeri  Recommended
3                    Kengeri  Recommended
4                    Kengeri  Recommended
...                      ...          ...
197425               Gangtok        Momos
197426               Gangtok        Momos
197427               Gangtok        Momos
197428               Gangtok        Momos
197429               Gangtok        Momos

                                            Dish Name  Price (INR)  \
0                                  Butter Murukku-200gm        133.9
1                                            Badam Milk         52.0
2                                       Chow Chow Bath        117.0
3                                          Kesari Bath         65.0
4                                           Mix Raitha        130.0
...                                                ...          ...
197425  Soya cheese chilli momo                    ...        112.0
197426  Kurkure momo fried                         ...        140.0
197427                              Chilli cheese momo        126.0
197428                                 Veg Momos (8 Pc)         85.0
197429                                          Soya Momo        100.0

        Rating  Rating Count
0          4.0            0
1          4.5           25
2          4.7           48
3          4.6           65
4          4.0            0
...        ...          ...
197425     4.4            0
197426     4.4            0
197427     4.4            0
197428     4.4            0
197429     4.4            0

[197430 rows x 10 columns]>
```

DATATYPES

In [24]:
```
df.dtypes
```

Out[24]:
```
State                      object
City                       object
Order Date          datetime64[ns]
Restaurant Name            object
Location                   object
Category                   object
Dish Name                  object
Price (INR)               float64
Rating                    float64
Rating Count                int64
dtype: object
```

In [25]:
```
df.describe()
```

Out[25]:

	Order Date	Price (INR)	Rating	Rating Count
count	197430	197430.000000	197430.000000	197430.000000
mean	2025-05-01 19:41:20.996808960	268.512920	4.341582	28.321805
min	2025-01-01 00:00:00	0.950000	1.500000	0.000000
25%	2025-03-01 00:00:00	139.000000	4.300000	0.000000
50%	2025-05-02 00:00:00	229.000000	4.400000	2.000000
75%	2025-07-01 00:00:00	329.000000	4.500000	15.000000
max	2025-08-31 00:00:00	8000.000000	5.000000	999.000000
std	NaN	219.338363	0.422585	87.542593

KPI's

Total Sales

In [26]:
```
df["Price (INR)"].sum()
```

Out[26]:
```
np.float64(53012505.769999996)
```

In [27]:
```
total_sales = df ["Price (INR)"].sum()
print ("Total Sales (INR):", total_sales)
```

```
Total Sales (INR): 53012505.769999996
```

Average Rating

In [33]:
```
average_rating  = df["Rating"].mean()
print("Average Rating (INR):", round(average_rating,2))
```

```
Average Rating (INR): 4.34
```

Average Order Value

In [36]:
```
avg_order_value = df["Price (INR)"].mean()
print ("Avg Order Value (INR):",round(avg_order_value,2))
```

```
Avg Order Value (INR): 268.51
```

Rating Count

In [37]:
```
rating_count = df["Rating Count"].sum()
print("Ratings Count:", round(ratings_count,2))
```

```
Ratings Count: 5591574
```

Total Orders

In [39]:
```
total_orders = len(df)
print("Total_Orders", round(total_orders,2))
```

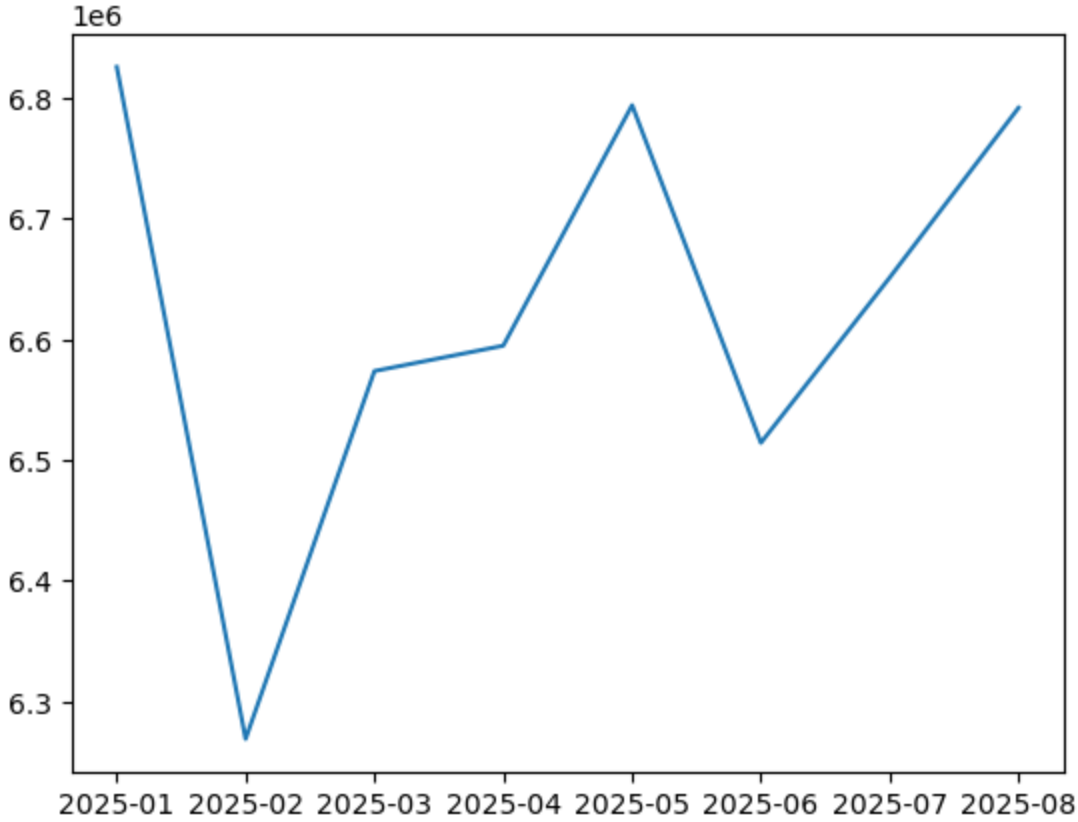
```
Total_Orders 197430
```

In [40]:
```
# Charts design
```

Monthly Sales Trend

In [45]:
```
df["Order Date"] = pd.to_datetime(df["Order Date"])

df["YearMonth"] = df["Order Date"].dt.to_period("M").astype(str)

monthly_revenue = df.groupby("YearMonth")["Price (INR)"].sum().reset_index()

plt.figure()
plt.plot(monthly_revenue["YearMonth"], monthly_revenue["Price (INR)"])
plt.xsticks(rotations=45)
plt.xlabel("Month")
plt.ylabel("Revenue (INR)")
plt.title("Monthly Revenue Trend")
plt.tight_layout()
plt.show()
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[45], line 9
      7 plt.figure()
      8 plt.plot(monthly_revenue["YearMonth"], monthly_revenue["Price (INR)"])
----> 9 plt.xsticks(rotations=45)
     10 plt.xlabel("Month")
     11 plt.ylabel("Revenue (INR)")

AttributeError: module 'matplotlib.pyplot' has no attribute 'xsticks'
```



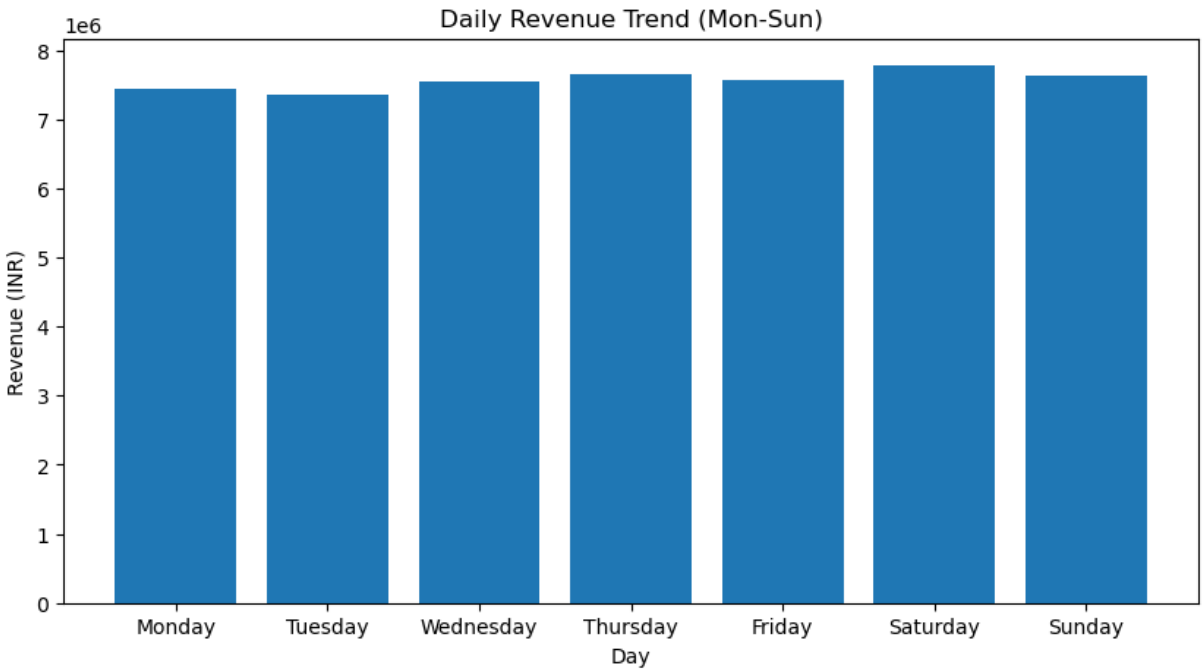
Daily Sales Trend

In [2]:
```
import pandas as pd
```

In [9]:
```
df["DayName"] = pd.to_datetime(df["Order Date"]).dt.day_name()

daily_revenue = (
    df.groupby("DayName")["Price (INR)"]
    .sum()
    .reindex(["Monday","Tuesday","Wednesday","Thursday","Friday","Saturday","Sunday
)

plt.figure(figsize=(10,5))
plt.bar(daily_revenue.index, daily_revenue.values)
plt.title("Daily Revenue Trend (Mon-Sun)")
plt.xlabel("Day")
plt.ylabel("Revenue (INR)")
```

```
plt.xticks(rotations30)

plt.show()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[9], line 14
     12 plt.xlabel("Day")
     13 plt.ylabel("Revenue (INR)")
---> 14 plt.xticks(rotations30)
     16 plt.show()

NameError: name 'rotations30' is not defined
```



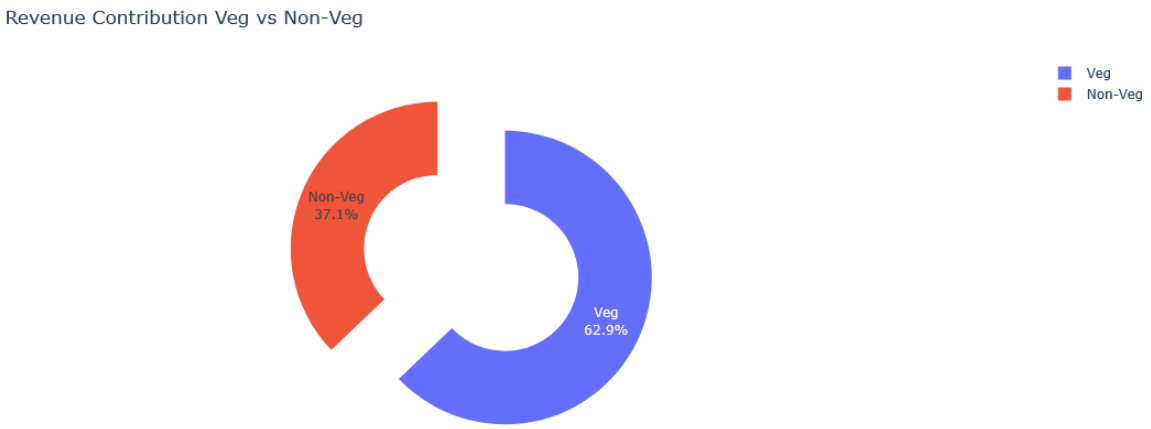
Total Sales by Food Type (Veg vs Non-Veg)

In [13]:
```
non_veg_keywords = [
    "chicken","egg","fish","mutton",
    "prawn","biryani","kabab","kebab",
    "non-veg", "non veg"
]

df["Food Category"] =np.where(
    df["Dish Name"].str.lower().str.contains("|".join(non_veg_keywords),na=False),
    "Non-Veg",
    "Veg"
)
```

In [14]:
```
food_revenue = (
    df.groupby("Food Category")["Price (INR)"]
    .sum()
    .reset_index()
```

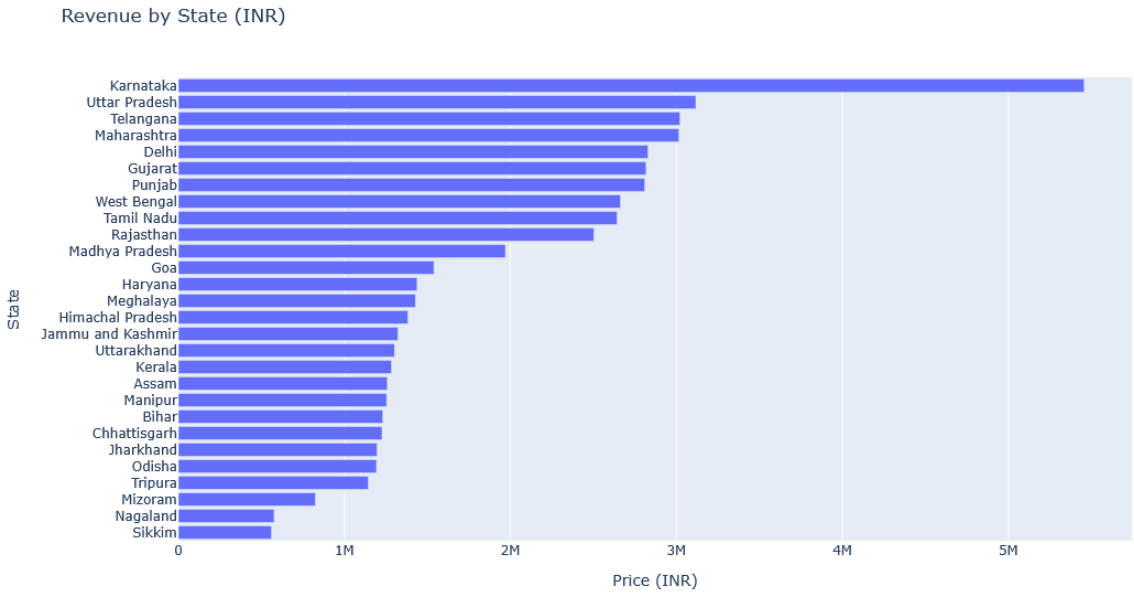
```
)
```

In [17]:
```
fig = px.pie(
    food_revenue,
    values="Price (INR)",
    names="Food Category",
    hole= 0.5,
    title = "Revenue Contribution Veg vs Non-Veg",
)

fig.update_traces(
    textinfo="percent+label",
    pull=(0.50,0)
)

fig.update_layout(
    height=500,
    margin=dict(t=60, b=40 ,l=40 ,r=40)
)
fig.show()
```



Total Sales by State

In [30]:
```
fig = px.bar(
    df.groupby("State", as_index=False)["Price (INR)"].sum()
      .sort_values("Price (INR)", ascending=False),
    x="Price (INR)",
    y="State",
    orientation="h",
    title="Revenue by State (INR)"
)

fig.update_layout(height=600, yaxis=dict(autorange="reversed"))
fig.show()
```



Quarter Performance Summary

df["Order_Date"] = pd.to_datetime(df["Order Date"]) df["Quarter"] = df["Order_Date"].dt.to_period("Q").astype(str) quarterly_summary = (df.groupby("Quarter", as_index=False) .agg(Total_Sales=("Price (INR)", "sum"), Avg_Rating=("Rating", "mean"), Total_Orders=("Order_Date", "count")) .sort_values("Quarter"))

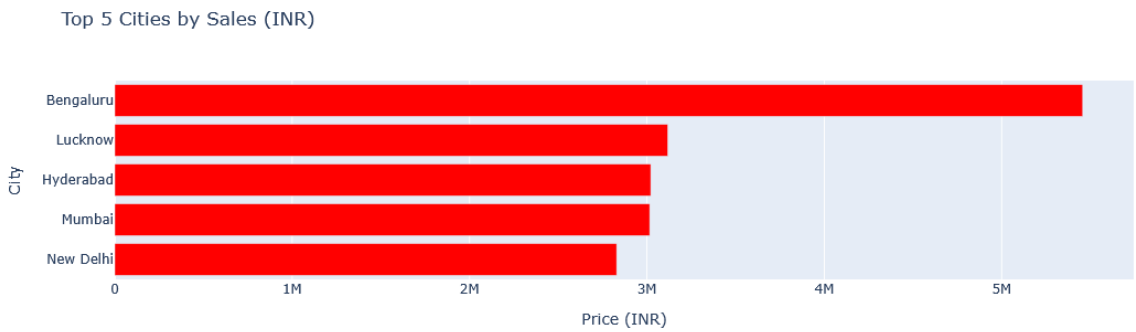
quarterly_summary["Total_Sales"] = quarterly_summary["Total_Sales"].round(0) quarterly_summary["Avg_Rating"] = quarterly_summary["Avg_Rating"].round(2)

quarterly_summary

Top 5 Cities by Sales

In [34]:
```
top_5_cities = (
    df.groupby("City")["Price (INR)"]
      .sum()
      .nlargest(5)
      .sort_values()
      .reset_index()
)

fig = px.bar(
    top_5_cities,
    x="Price (INR)",
    y="City",
    orientation="h",
    title="Top 5 Cities by Sales (INR)",
    color_discrete_sequence=["red"]
)
```

```
fig.show()
```


Top 5 Cities by Sales (INR)
Bengaluru
Lucknow
Hyderabad
Mumbai
New Delhi
City
0
1M
2M
3M
4M
5M
Price (INR)

In [ ]: